20 Two point charges of the same sign are placed 4.0 m apart. The magnitude of the charges is in the ratio 2:1. The potential at the mid-point between the two charges is 6740 V.

If the charge of smaller magnitude is changed to a negative charge of the same magnitude, calculate the electric field strength at the mid-point between the two charges.

A 6740 N C^{-1} **B** 3370 N C^{-1} **C** 1690 N C^{-1} **D** 1120 N C^{-1}